

Application of Newton's Method: Interior Point Method

Recall: Newton's method in quad conv. regime - "Phase II"
requires $\log \log(1/\varepsilon)$ iterations to obtain ε -accurate
solution.

Thus far: $\min: f(x)$

Linear Optimization

$$\begin{array}{ll} \min: & c^T x \\ \text{st:} & Ax \leq b \end{array}$$

$\longleftrightarrow$

$$\begin{array}{ll} \min: & c^T x \\ \text{st:} & Ax = b \\ & x \geq 0 \end{array}$$

Applications

Resource allocation

Scheduling

Routing

... many combinatorial opt'n problems,

(a) $\hat{x}$ an interior point

(b) $\hat{x}$ a corner point.

Optimality conditions

$$\hat{x} \text{ opt sol'n: } 0 \in \partial f(\hat{x}) + N_{\mathcal{X}}(\hat{x})$$

$$0 \in c + N_{\mathcal{X}}(\hat{x})$$

Exercise: write down the normal cone at $\hat{x}$

Linear Optimization

Why is LP so powerful?

① If a soln exists, there is always a soln at an ext point

Number of extreme points $\longleftrightarrow$ ② Many corners.

$$\mathcal{P} = \{x: Ax \leq b\} \quad m \times n \quad m \text{ inequalities in } n \text{ dims}$$

corners $\sim \exp(m)$ (as big as)

simple example: $[0,1]^n$ $m=2n$, 2^n corners

Challenge: Find a polytope with $\exp$ # facets in # vertices
i.e., $m \approx 2^n$ corners

Simplex Method

① Projected Gradient Descent

main cost: projection step. $\rightarrow$

$$\min: \|z - x\|_2^2$$

$$\text{st: } Ax \leq b$$

② Simplex: searches corner to corner

Interior Point Method for LP

Idea: avoid corners

$$\min: c^T x$$

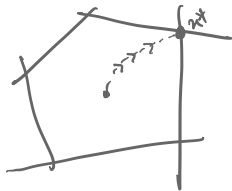
$$\text{st: } Ax \leq b$$

$$\rightarrow a_i^T x \leq b_i, i=1, \dots, m$$

$$\min: F_t(x) = \boxed{t \cdot c^T x} - \underbrace{\sum_{i=1}^m \log(b_i - a_i^T x)}_{\substack{\text{Barrier} \\ \downarrow \\ \text{as } a_i^T x \rightarrow b_i \\ -\log(b_i - a_i^T x) \rightarrow +\infty}}$$

Does not allow us to approach barrier.

$$\text{As } t \rightarrow \infty, \quad c^T x_t \rightarrow c^T x^*$$



At each ~~step~~ iteration we solve: $\min: F_t(x)$

Semidefinite Optimization

$$A, B \in S^n$$

$$A \in S_+^n \Leftrightarrow \lambda_i(A) \geq 0$$

" $A \succeq 0$ "

Induces a natural ordering:

$$A \preceq B \Leftrightarrow (A - B) \preceq 0$$

$$\Leftrightarrow (A - B) \in S_+^n$$

SDP: min: $\sum c_i x_i$ (scalar) $\leftarrow$ $n \times n$ $\leftarrow$ $n \times n$

st: $\sum A_j x_j \preceq B$

$$LP: \sum A_j x_j \leq b$$

$$Ax \leq b$$

$$b - Ax \geq 0$$

$$(b - Ax) \in \mathbb{R}_+^n$$

min: $\sum c_{ij} x_{ij}$ (scalar) $\leftarrow$ $n \times n$ $\leftarrow$ $n \times n$ $\leftarrow$ $\text{Tr}(C^T X)$

st: $\text{Tr}(A^T X) = b$ (scalar) $\leftarrow$ $n \times n$

LP: Barrier for $\mathbb{R}_+^n$: $-\sum \log x_i$

Barrier for S_+^n : $-\sum \log \lambda_i(X) = -\log \prod \lambda_i(X) = -\log \det X$

$X \succeq 0$

Interior Point Method for Semidefinite Optimization

$$\text{SDP: } \min: \sum c_i x_i$$

$$\text{st: } B - \sum A_i x_i \succeq 0$$

$$F_t(x) = t \cdot c^T x - \log \det (B - \sum A_i x_i)$$

$$\text{IPM: } x_t = \operatorname{argmin}: F_t(x)$$

$$\text{let } t \rightarrow \infty$$

Agenda: applying Newton's Method

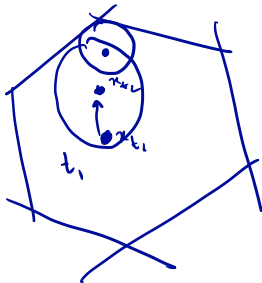
For LP:

$$\begin{aligned} \min: & c^T x \\ \text{st: } & Ax \leq b \end{aligned}$$



$$\min: F_t(x) = t \cdot c^T x - \sum \log(b_i - a_i^T x)$$

Need: ① Need to be able to solve $\min: F_t(x)$ quickly. (quad. conv. via Newton)



② Increase t quickly: $t^+ = \alpha t$

③ Next $c^T x_t$ to be close to $c^T x^*$